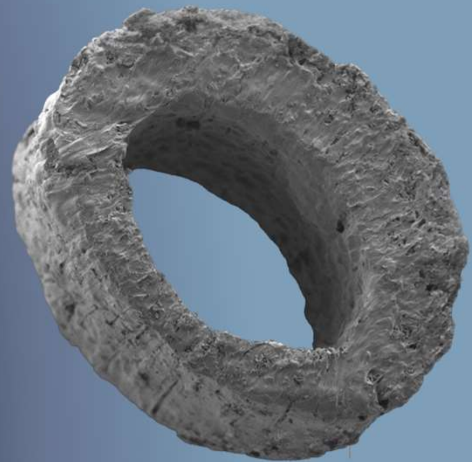
THE SUNDAY TIMES
Best Places to Work 2024
SMALL ORGANISATION

INVESTORS IN PEOPLE®
We invest in people Platinum

INVESTORS IN PEOPLE®
We invest in wellbeing Platinum

Advances in Interbody Cages for Spinal Fusion



LocateBio
Reshaping rhBMP-2 Perspectives

The challenge

Design a cage concept with an adaptive modulus to allow the new bone that has grown into the pores of the cage to bear an increasing load over time.

My top 3 design considerations

1. Patient Safety: the ability to bear load until it is certain that new bone will have formed. i.e. aligning temporal adaptive modulus with bone formation - but be cautious.
2. Include design features that drive bone formation
3. Ability to assess bone formation (radiolucency)

Assessing bone formation

LocateBio

3. Ability to assess bone formation (radiolucency)

An example of the issues it can cause when radiopaque

Research Article

BIOMEDICAL
Journal of Scientific & Technical Research

ISSN: 2574 -1241

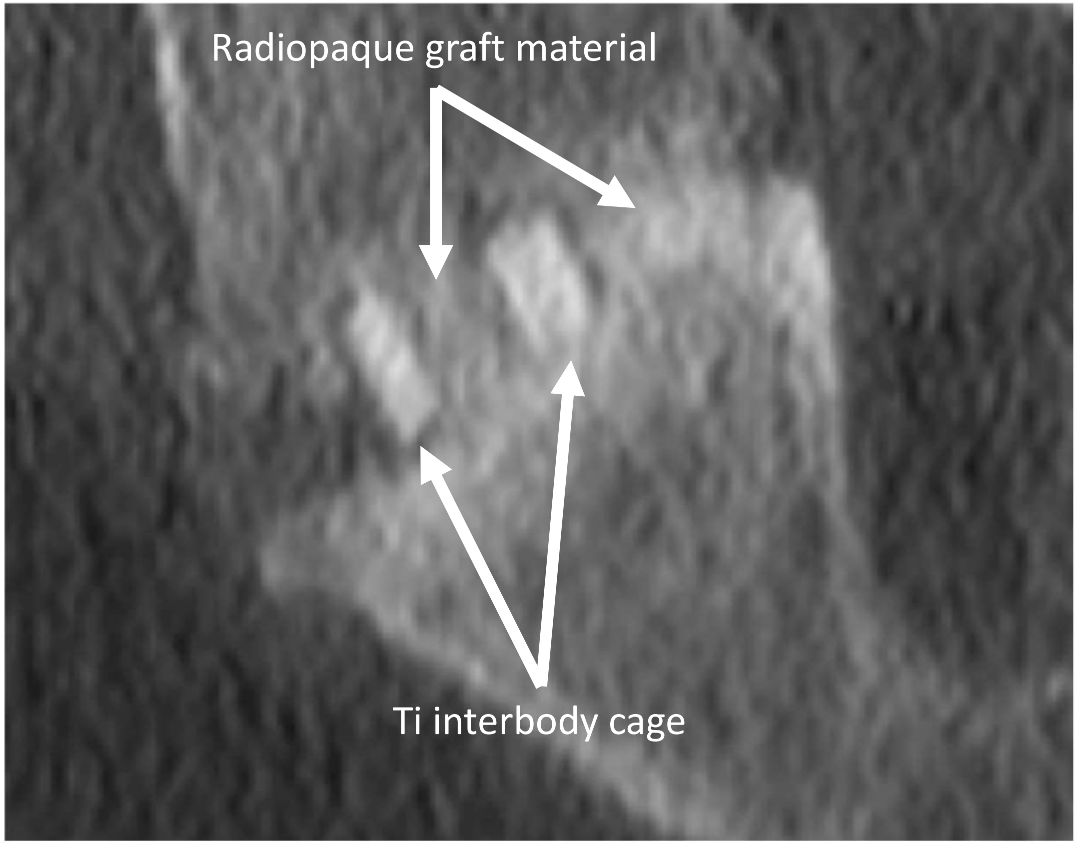
DOI: 10.26717/BJSTR.2024.54.008570

First-In-Human Study with a Novel Synthetic Bone Graft, OssDsign Catalyst™, in Transforaminal Lumbar Interbody Fusion with Instrumented Posterolateral Fusion

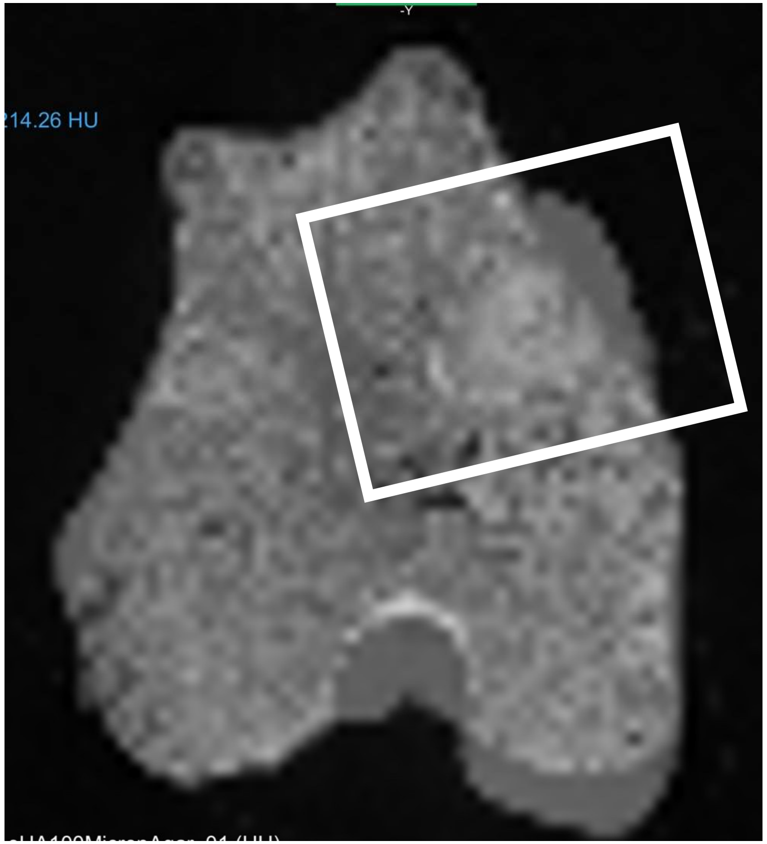
- Reports a 93% fusion rate, which is very high.
- Ceramic (Si-HA) only material (osteoconductive)

Human vs Sheep Defect Study (8x15mm)

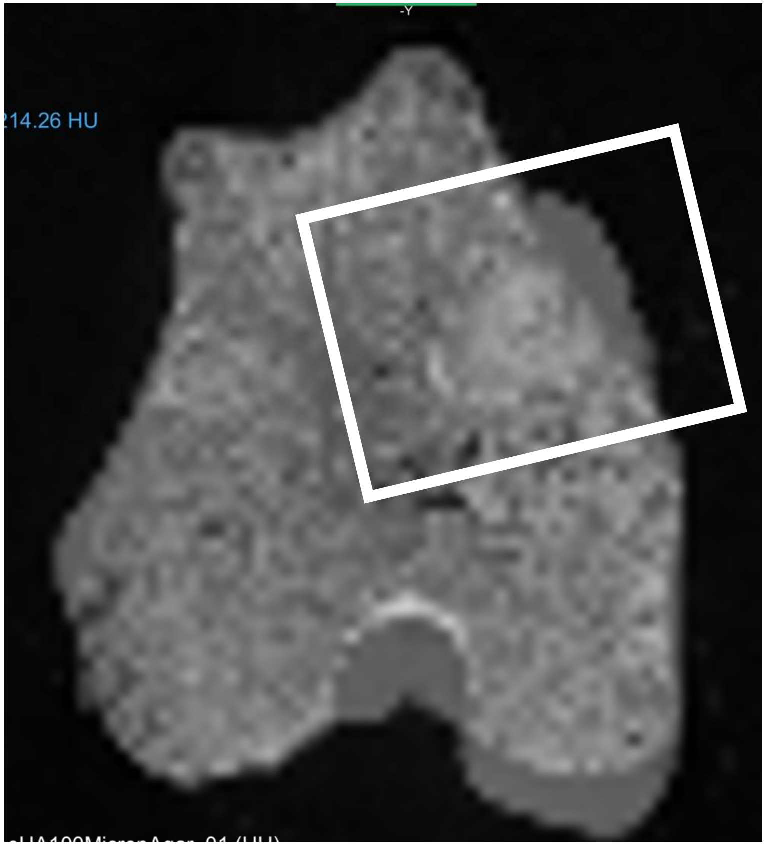
Data from their human clinical study

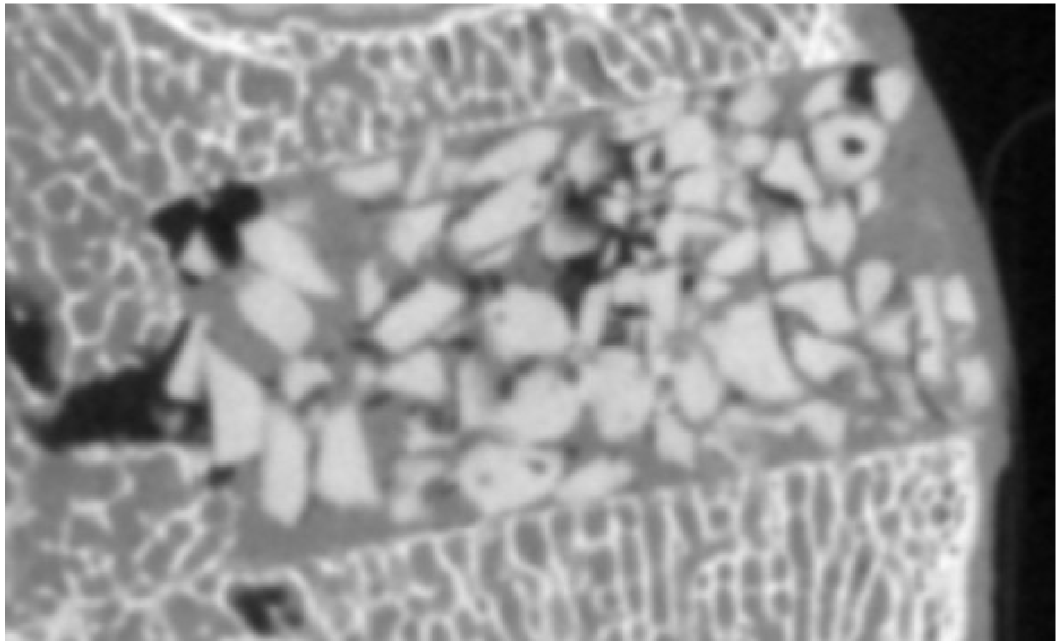


Data from our preclinical study



A very challenging model (sheep cadaver)



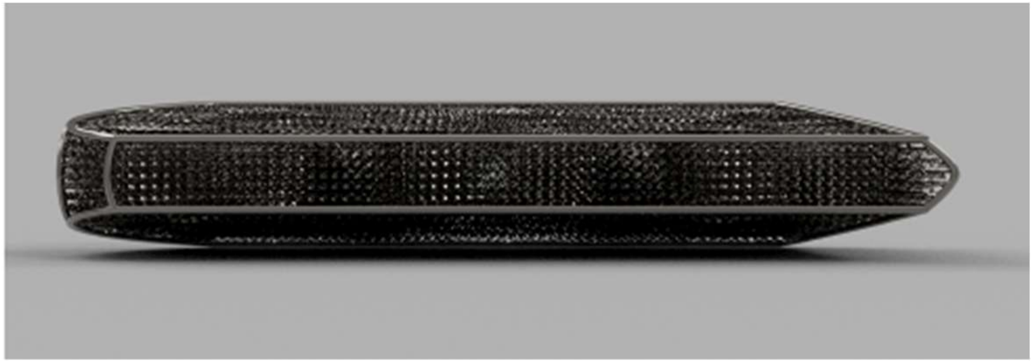


Going further

✓All groups recognised the importance of radiolucency.

✓Some prioritised all polymer-based materials over Ti or other radiopaque materials – but polymers have some disadvantages (see later section on patient safety).

✓Adaptive radiolucent: For example - Group 20’s design would be very radiopaque initially, but the resorbing elements would allow increase visualisation of new bone formation over time.



LocateBio

Contributing to bone formation

2. Contribution to bone formation

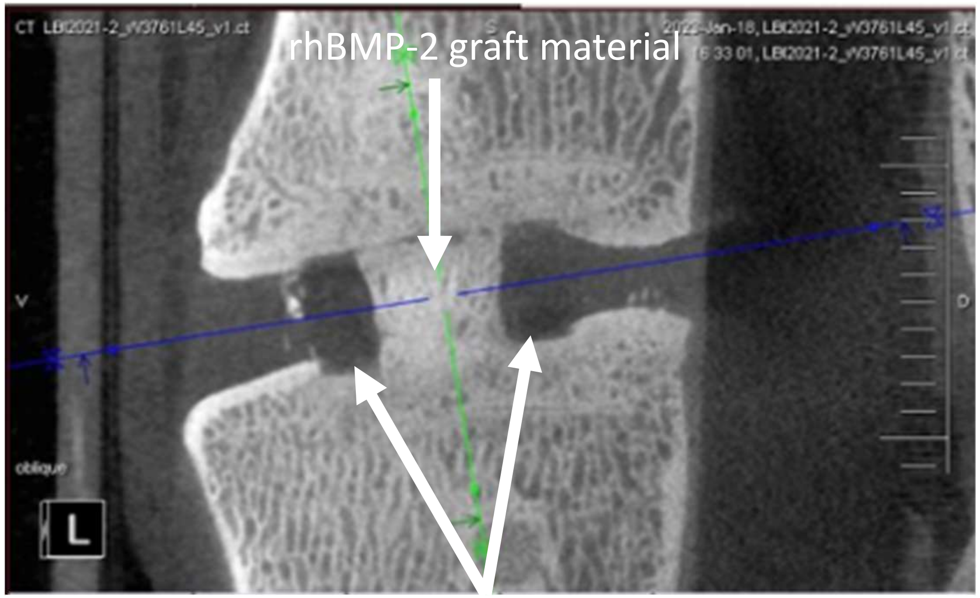
Surface features considered by the groups

✓ Pores (focus on size – waste/nutrients/blood vessels/bone)

✓ Bone on-growth e.g. Ti vs PEEK (addressed with HA coating)

• rhBMP-2 coating

An alternative: The use of nanoscale Ti as a coating for PEEK

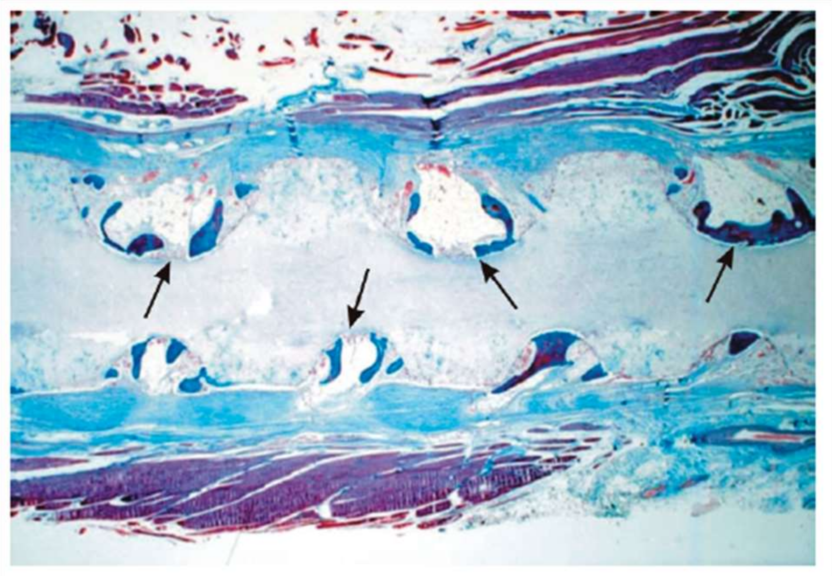


Ti coated PEEK Cage

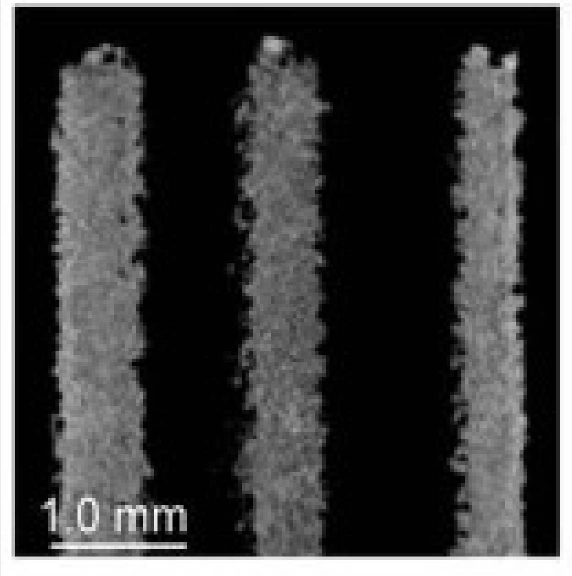
LocateBio

Going 1 step further – shapes and surface

Macro Scale Matters



Ripamonti U., J. Cell. Mol. Med. Vol 8, No 2, 2004 pp. 169-180

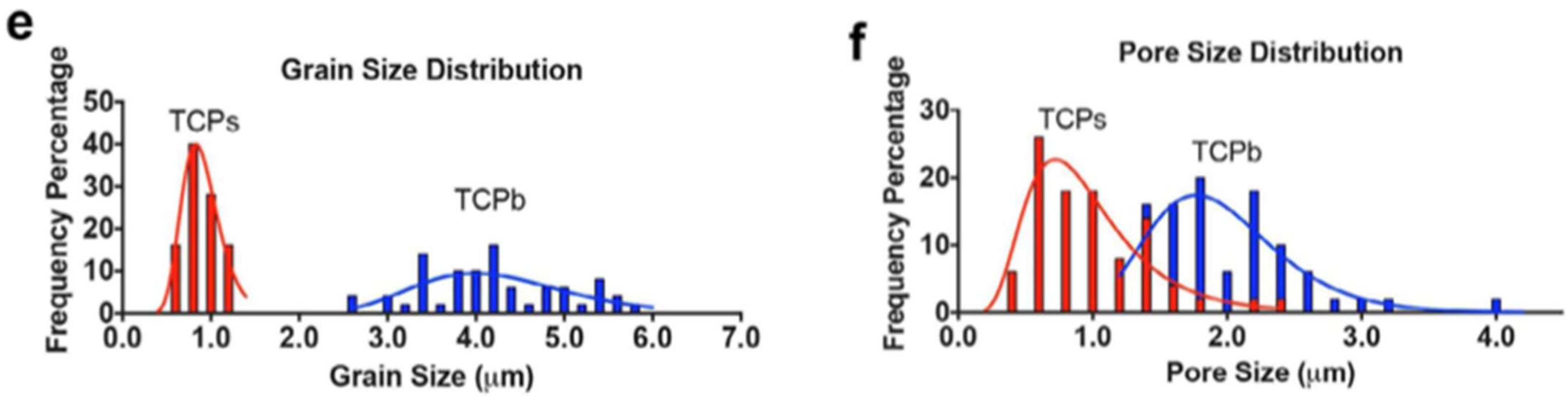


A. Fukuda et al., Acta Biomater. 7 (5) (2011) 2327

LocateBio

Insight driven design

Nano-scale matters



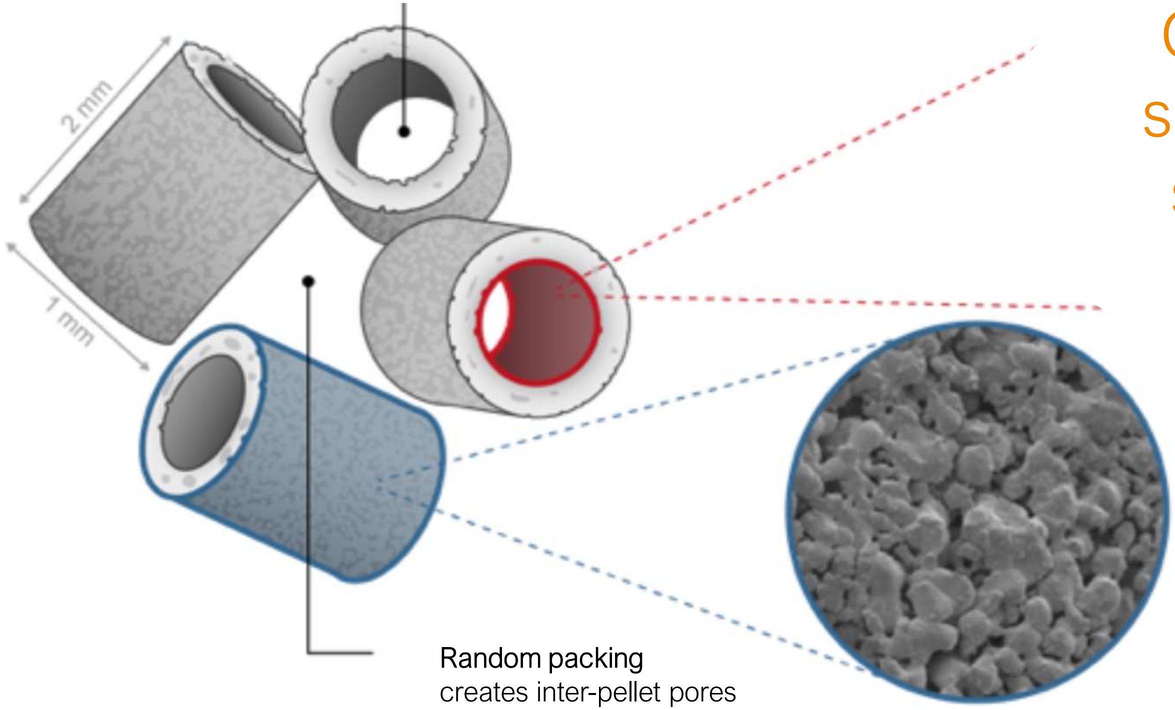
LocateBio

https://www.researchgate.net/profile/Noel-Davison/publication/261736686_Submicron-scale_surface_architecture_of_tricalcium_phosphate_directs_osteogenesis_in_vitro_and_in_vivo/links/543e7daa0cf2e76f02228187/Submicron-scale-surface-architecture-of-tricalcium-phosphate-directs-osteogenesis-in-vitro-and-in-vivo.pdf

Insight driven design

In Vitro Testing

Pellet inner lumen
provides a macro pore for angiogenic ingrowth, and a 'smart concavity' microenvironment



Random packing
creates inter-pellet pores

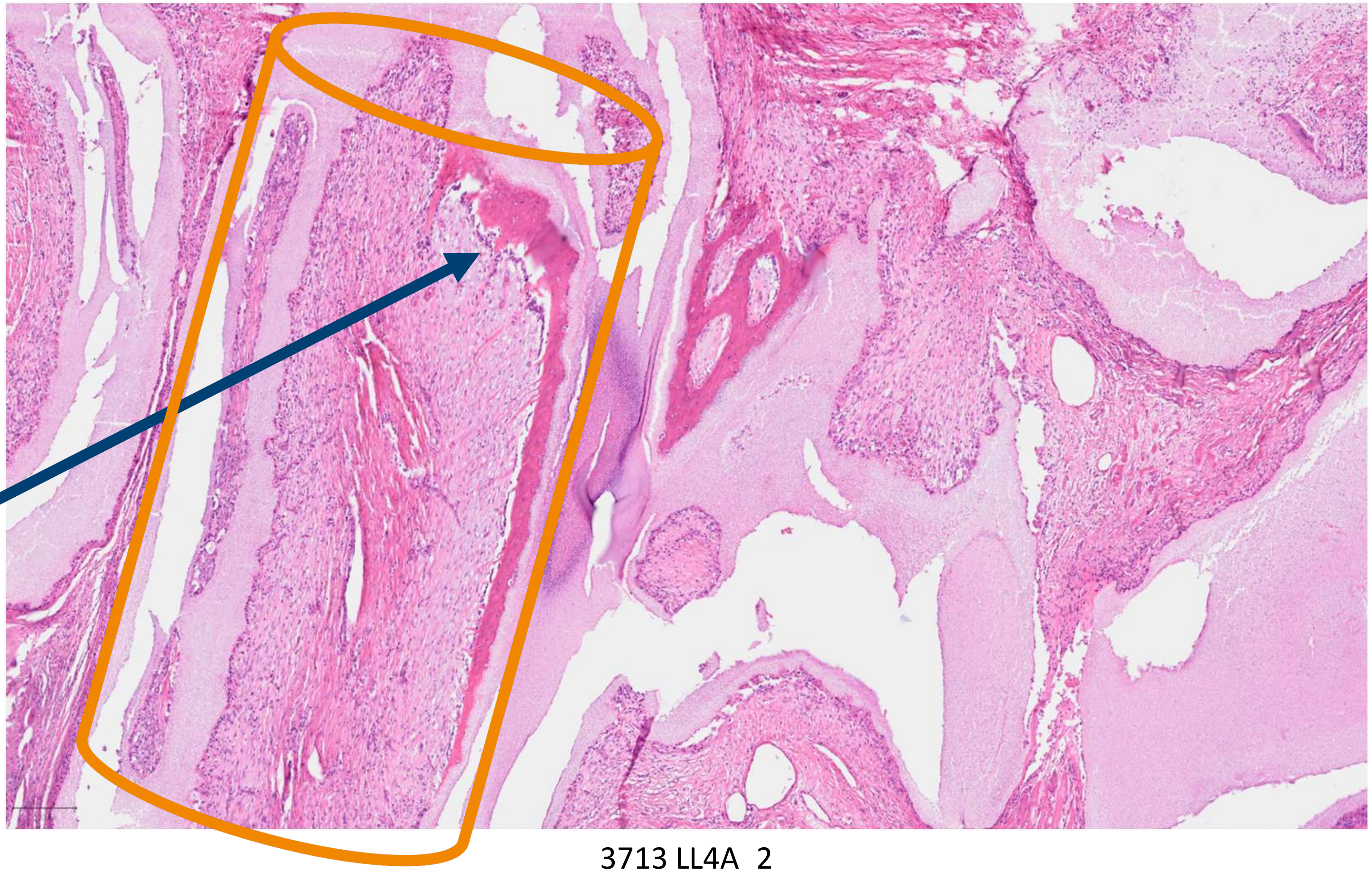
Combines concave surfaces (tubes) and submicron surface features

LocateBio

Testing macro features only (shape)

Intramuscular implantation – no osteoconduction, any new bone would be osteoinduction

New bone at 6 weeks



3713 LL4A_2

Testing macro features only (shape)

12 weeks

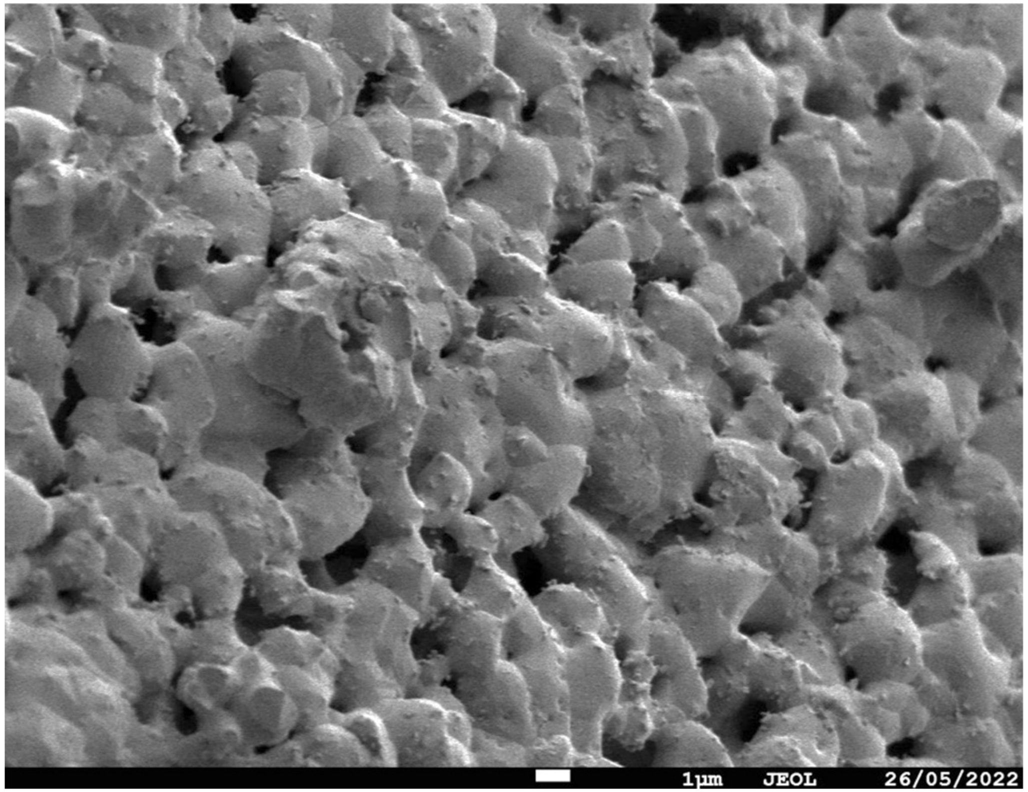
- Significant new bone at 12 weeks

Osteoinduction from a ceramic based just on a concave shape!

SEM – different surface topographies

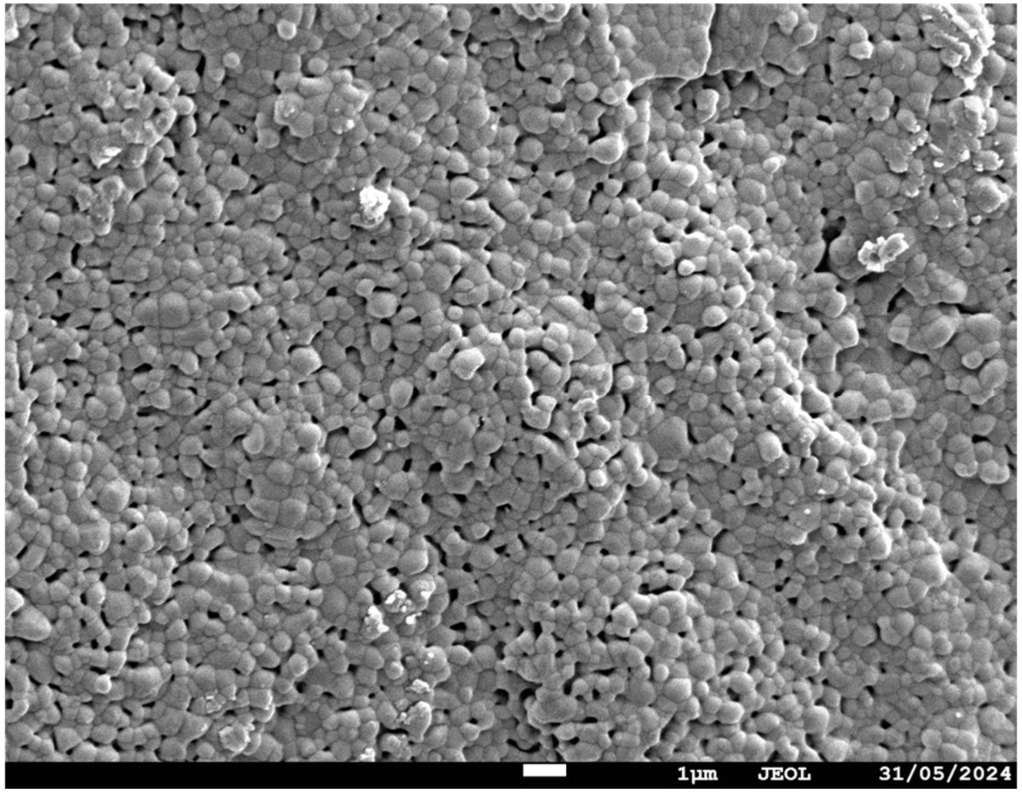
x4000

Non-submicron features



x5000

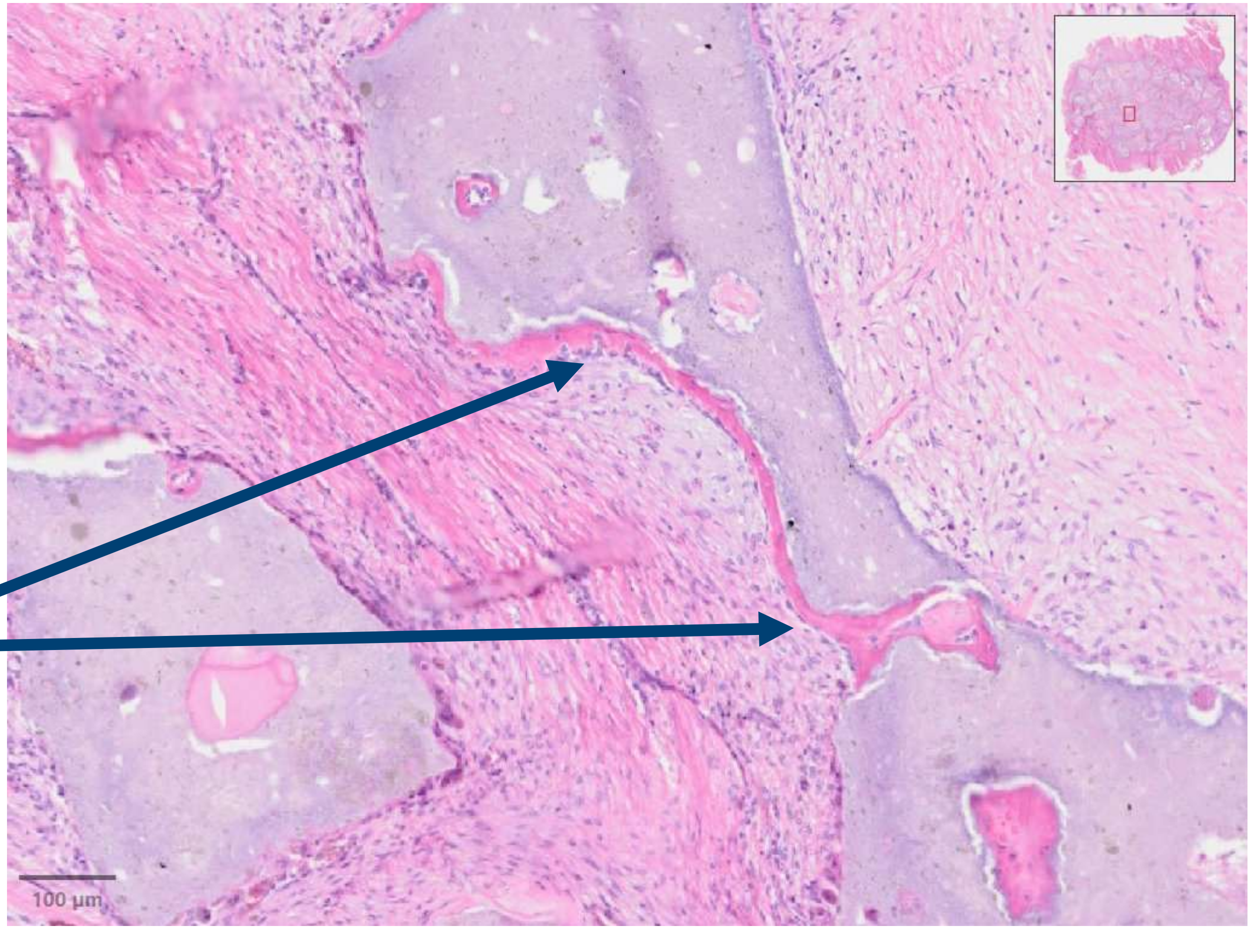
Submicron features



Testing nano-scale features only

Osteoinduction from a ceramic based just on a nano-scale surface features!

New bone at 6 weeks



LBI2024-1 3878 LL1b H&E - 20x_BF_01-3

Going further with your deigns

Think about

- Nanoscale features on the surface of your cages (easier to do with Ti)
- Not just size of your pores, but also their shapes (concave) – lots of options with 3D printing
- Add a permanent surface charge – acts to attract and bind endogenous growth factors to the surface
- Consider Piezoelectric materials (e.g. polyvinylidene fluoride – PVDF)

LocateBio

Patient Safety

1. Patient Safety

Polymer only cages….

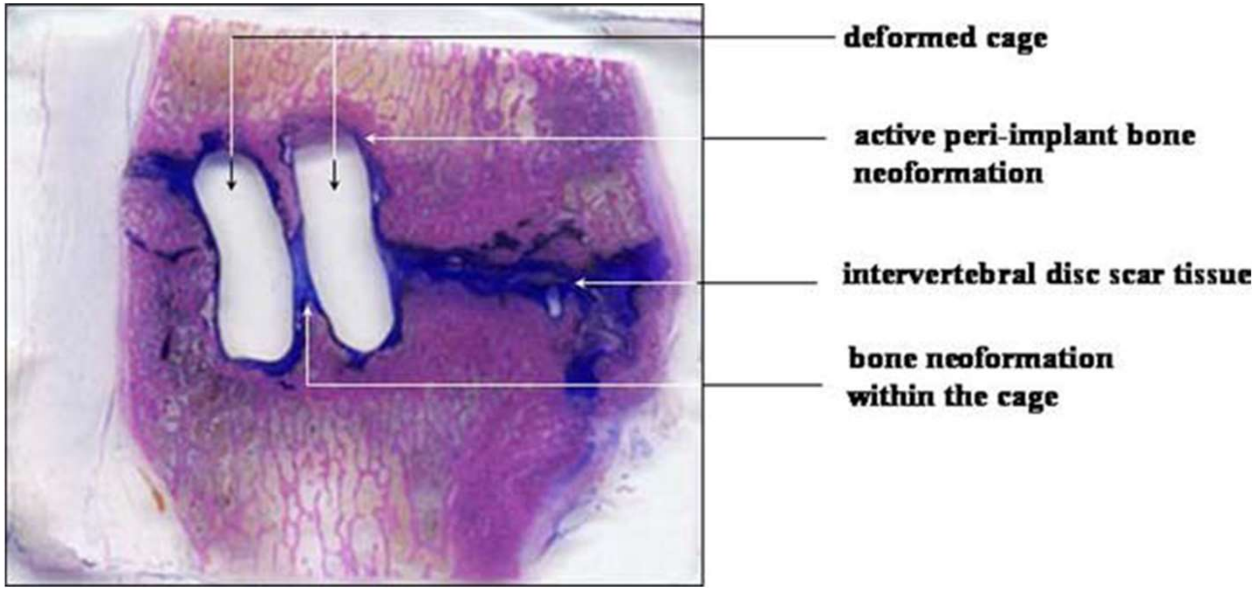


Figure 1 - Lazennec et al. 2006

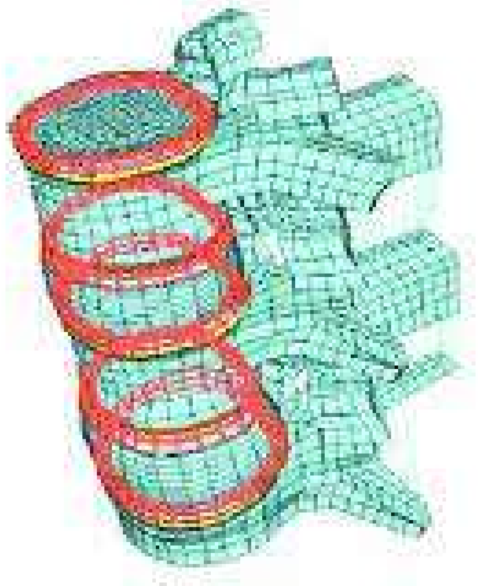
Figure 2 – Toth et al (2002)

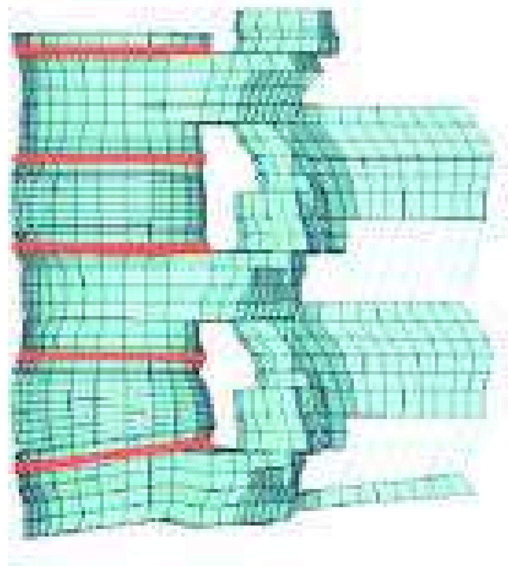
LocateBio

1. Patient Safety (continued)

Apophseal Ring

- Apophseal Ring bears the most weight – very strong cortical bone.
- The centre of the vertebral body contains the softer, cancellous bone
- Some groups had the resorbable parts of the cage on the outside and the harder, permanent parts in the inside.
- It needs to be the other way around





The Best Bits….

- ✓ The use of 2 materials to allow complementary aspects to overcome the weakness of just 1 material
  - Ti or Si_3N_4 or Coated PEEK as the permanent outer layer
  - Use of innovate resorbable materials for the inner (Mg Alloys)
- ✓ Use of i) porosity and ii) innovate lattice structures (e.g. TPMS lattices) to offset modulus issues with harder materials like Ti.
- ✓ Coating to slow Mg degradation (as likely too quick for new bone growth)
- ✓ Use of antimicrobial coatings (Silver)

LocateBio

Some 'builds' on your best bits

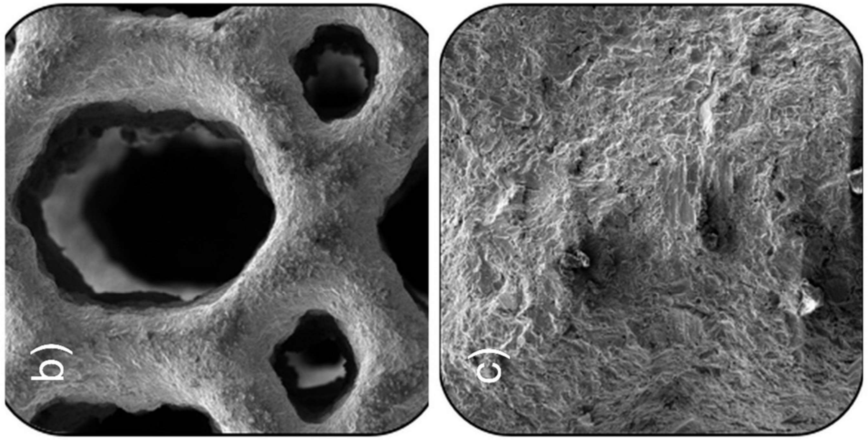
- Copper vs Silver for antimicrobial coating?
  - Copper mimics a hypoxic signal and leads to upregulated angiogenesis (blood vessels are important for bone formation)

- Slot-in design for a personalised resorbable inner?
  - Could have a fast, medium and slow resorbing inner, and the surgeon could assess the bone quality during surgery and select a more patient specific resorption profile vs one-size-fits-all resorption rate.

LocateBio

My design

- 3D printed Ti outer to sit on the Apophseal Ring
- Open, porous structure with concave pores and submicron surface features
- A slot-in resorbable inner made, at least in part, from a Mg alloy to allow personalisation at point of use (fast, medium and slow)…likely to be coated (chitosan)
- In the very centre, a bone graft chamber for point of use addition of bone graft substitutes (e.g. rhBMP-2)



Summary

- Chronic low back pain
  - World Health Organisation – leading cause of disability worldwide
  - Debilitating, socially isolating

- Biomedical engineering has the potential to significantly improve outcomes for patients, and change lives.

LocateBio